

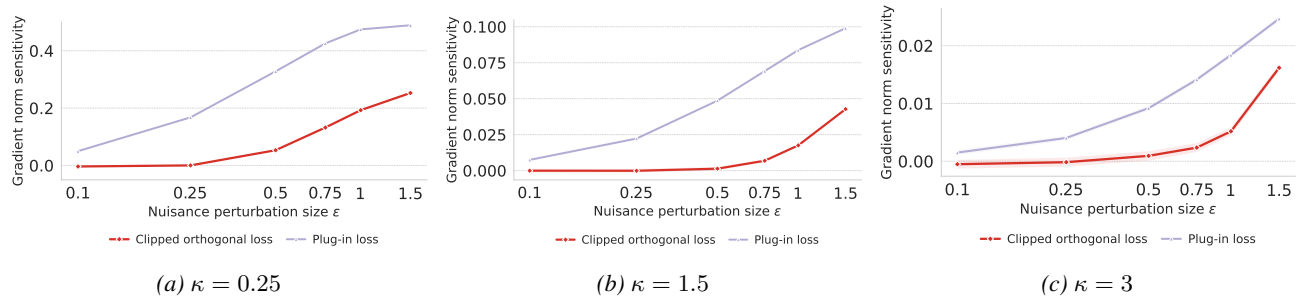


Figure 2. Empirical orthogonality diagnostic. Sensitivity of the parameter-gradient norm of the training objectives to nuisance perturbations around the true nuisances. Across representative values of κ , the clipped orthogonal objective exhibits substantially lower local sensitivity than the plug-in objective. *Y-axis not aligned for visual clarity.*